

VendorGuard AI – Autonomous Financial Reconciliation

A governed, multi-agent financial firewall preventing enterprise billing leakage through strict PO grounding

3% – 7%

SILENT BILLING LEAKAGE
INTERCEPTED

Dual Ingest

PDF UPLOADS + DIRECT JSON
FEEDS

3 Isolated

SPECIALIZED AGENT ROLES

₹0.68

DIRECT COST PER AUDIT RUN

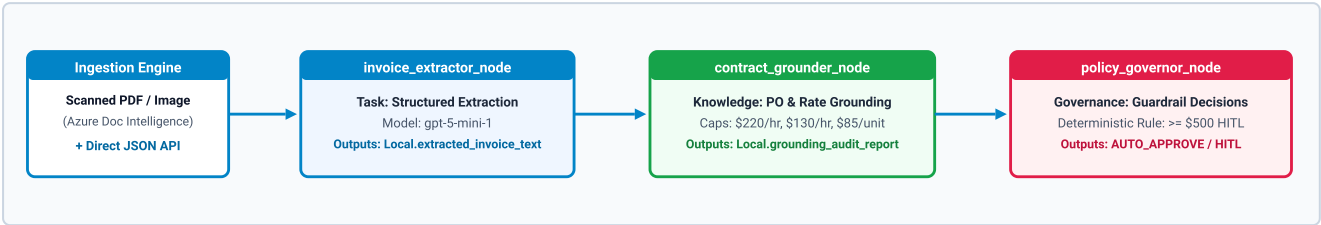
1. The Problem We Set Out to Solve

Enterprises lose **3% to 7% of procurement spend** to billing leakage: rates creep \$20–\$40 above agreed caps, contractors bill unauthorized roles, or expired POs get reused. Accounts Payable teams face thousands of manual invoices weekly and rubber-stamp payments to clear backlogs.

Why single-prompt prototypes fail: Asking a single LLM to extract messy data, cross-reference contract rate cards, do arithmetic, and decide payments leads directly to hallucinations. In financial systems, financial guardrails must be deterministic, mathematically grounded, and split into specialized concerns.

2. Dual Ingestion & Sequential Workflow Topology

In production, reconciliation does not assume pristine JSON. Our ingestion layer supports **scanned PDF uploads** (via Azure AI Document Intelligence prebuilt invoice models or multimodal vision) alongside direct ERP JSON payloads, emitting normalized data into our 3-node Foundry workflow:



3. Specialization & Dataflow Contracts

Pipeline Stage	Archetype	Grounding Source	Operational Duty
PDF / JSON Ingest	Ingestion Layer	Document Intelligence / API Feed	Extracts key-value pairs and tabular line items from PDF scans or JSON into standard schema.
invoice_extractor_node	Task Specialist	Sanitized Inbound Payload	Standardizes vendor ID, PO number, hours, and rates into strict JSON without guessing validity.
contract_grounder_node	Knowledge Specialist	Corporate PO & Rate Registry	Performs exact mathematical audits against contracted rate caps. Flags line variances.
policy_governor_node	Governance Specialist	Disbursement Threshold Rules	Applies corporate policy: triggers straight-through disbursement or halts payment for human sign-off.

4. Deterministic Financial Guardrails

Financial reconciliation cannot rely on stochastic LLM judgments. The policy node enforces hard business logic:

- Rule 1 – Zero Friction:** If **Variance = \$0.00** & PO matches → **AUTO_APPROVE** (HITL: false). Direct automated AP disbursement.
- Rule 2 – Leakage Interception:** If **Variance ≥ \$500.00** → **STAGE_DISPUTE_HITL** (HITL: true). Locks payment; packages line-item dispute brief for human review.
- Rule 3 – Zero Trust Block:** If PO is unregistered or vendor is unapproved → **REJECT_UNAUTHORIZED**. Hard freeze before capital moves.

5. Verified Benchmark Evaluation Scenarios

Scenario	Vendor & PO	Audited Discrepancy Findings	Workflow Verdict	Human Review?
INV-101 Clean Run	CloudScale Logistics PO-2026-8801	SRV-LOG-01: Billed \$85/unit vs \$85 cap. Variance: \$0.00 . Verified match.	AUTO_APPROVE	No (Straight-thru)
INV-102 Leakage Flag	Apex Cloud Consulting PO-2026-9140	CON-AI-01: Billed \$260/hr vs \$220 cap across 40 hrs. Overbill: +\$1,600.00 .	STAGE_DISPUTE_HITL	Yes (Flagged)
INV-103 Security Check	Shadow Corp Global PO-FAKE-0000	Fake PO number, unverified vendor. Zero-trust security tripwire triggered.	REJECT_UNAUTHORIZED	No (Hard Stop)

6. Production Observability & Telemetry (Application Insights)

Telemetry Metric	Empirical Log (INV-101 / INV-102)	Production Impact
Total Latency	14.957s (INV-101) / 20.224s (INV-102)	Rapid multi-agent traversal across 3 hops without bottlenecking AP pipelines.
Token Accounting	21,383 In / 1,050 Out • 15,940 In / 1,722 Out	Decoupled schema nodes prevent prompt bloat and runaway token consumption.
Unit Economics	₹0.68 per invoice run (~\$0.008 USD)	Replaces \$5–\$15 manual audit labor per invoice with 99% cost reduction.
Security Identity	DefaultAzureCredential() (Azure RBAC)	Zero API keys or plaintext secrets in codebase; enterprise zero-trust compliant.

7. Verified Repository Deliverables

File Path	Artifact Type	Evaluator Verification Value
vendorguard_workflow.yaml	Declarative Schema (IaC)	Verifies agent node orchestration, trigger bindings, and PowerFx routing.
workflow_runner.py	Python SDK (2.1+)	Programmatic execution runner with streaming responses and trace logging.
test_payloads.json	Evaluation Fixtures	Contains benchmark scenarios INV-101, INV-102, and INV-103.

Summary for Judges: VendorGuard AI solves real procurement leakage by combining dual-mode ingestion (PDF OCR/Vision + JSON) with a 3-stage Foundry workflow. With deterministic \$500 threshold gating, strict contract grounding, and Application Insights observability, it delivers zero hallucinations at ₹0.68 per run.